Integration

Integration by substitution

1 Find $\int 2x(x^2+1)^4\,dx$ using the substitution $u = x^2 + 1$.

2 Find $\int x\sqrt{x^2+9}\,dx$ using the substitution $u = x^2 + 9$.

3 Find $\int (2x+1)(x^2+x)^3\,dx$ using the substitution $u = x^2 + x$.

4 Find $\int 3x^2(x^3-2)^4\,dx$ using the substitution $u = x^3 - 2$.

Integration by parts

5 Find $\int xe^x\,dx$ using integration by parts.

6 Find $\int x\cos x\,dx$ using integration by parts.

7 Find $\int \ln x\,dx$ using integration by parts with $u = \ln x$ and $dv = dx$.

8 Find $\int x^2e^x\,dx$ using integration by parts twice.

Integration using partial fractions

9 Find $\int \frac{5x+7}{(x+1)(x+2)}\,dx$, first decomposing into partial fractions.

10 Find $\int \frac{1}{x^2-4}\,dx$ using partial fractions, factoring $x^2 - 4 = (x-2)(x+2)$.

11 Find $\int \frac{3x-1}{x(x-3)}\,dx$, first decomposing into partial fractions.

12 Find $\int \frac{2x+3}{(x-1)(x+4)}\,dx$, first decomposing into partial fractions.

Integrating trigonometric, exponential and logarithmic functions

13 Find $\int \sin(3x)\,dx$ and $\int e^{2x}\,dx$.

14 Find $\int \frac{4x}{x^2+1}\,dx$.

15 Find $\int \tan x\,dx$, using the substitution $u = \cos x$.

16 Find $\int \frac{1}{2x+3}\,dx$.

Definite integrals using substitution

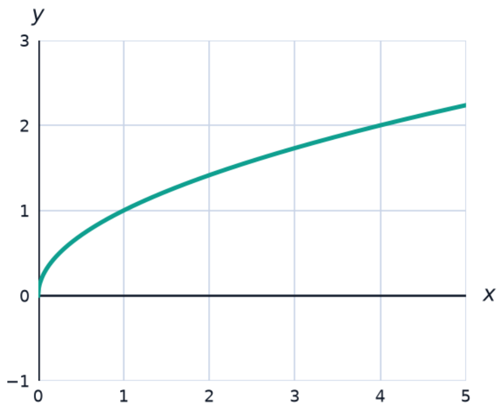
17 Evaluate $\int_0^2 x(x^2+1)^3\,dx$ using the substitution $u = x^2 + 1$.

18 Evaluate $\int_0^{\pi/2} \sin x \cos^3 x\,dx$ using the substitution $u = \cos x$.

19 Evaluate $\int_1^e \frac{\ln x}{x}\,dx$ using the substitution $u = \ln x$.

Volumes of revolution: the disk method

20 The region under $y = \sqrt{x}$ on $[0, 4]$ is revolved about the x-axis. Find the volume generated, using $V = \pi \int y^2\,dx$.



21 The region under $y = x^2$ on $[0, 2]$ is revolved about the x-axis. Find the volume.

22 The region under $y = e^x$ on $[0, 1]$ is revolved about the x-axis. Find the volume, using $V = \pi \int_0^1 e^{2x}\,dx$.

Volumes of revolution: washer and shell methods

23 The region between $y = x^2$ and $y = 9$ is revolved about the x-axis. Using the washer method, set up and evaluate the volume integral.

24 The region under $y = x^3$ on $[0, 2]$ is revolved about the y-axis. Use the shell method, $V = 2\pi \int_0^2 x \cdot x^3\,dx$, to find the volume.

A well-designed word problem: designing a vase

25 This question has three parts. A vase is formed by rotating the curve $y = \sqrt{x}$, for $0 \le x \le 9$ (cm), about the y-axis, where x is horizontal distance from the axis and y is height. (a) Express x^2 in terms of y, and evaluate $V = \pi\int_0^3 x^2\,dy$ to find the volume of the vase. (b) A cylindrical base of radius 3 cm and height 2 cm is attached underneath. Find the total volume, to two decimal places (using $\pi \approx 3.14159$). (c) Show that the volume of water filled to height h (ignoring the base) is $V(h) = \frac{\pi h^5}{5}$, and find the height h at which the water volume equals half the vase's volume from part (a), to two decimal places.